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Unit – 1 : Session – 8 : SLO – 1 | Asymptotic Notations — Big O, Big Omega, Big Theta

■ I. Fill in the Blanks

1. Big O — the upper bound on running time.
2. Big Omega (Ω) — the lower bound.
3. Big Theta (Θ) — the tight bound.
4. worst — Big O analyzes worst-case performance.
5. best — Big Omega analyzes best-case performance.
6. upper and lower — Θ bounds from both sides.
7. $O(n)$ — linear search.
8. $O(\log n)$ — binary search.
9. $O(n^2)$ — bubble sort worst case.
10. O — $f(n) = 3n + 2$ is $O(n)$.
11. O — $f(n) = 5n^2 + 3n + 7$ is $O(n^2)$.
12. lower-order — constants and lower-order terms are ignored.
13. $O(1)$ (constant) — same time regardless of input size.
14. Ω — $f(n) \geq c \cdot g(n)$ for $n \geq n_0$ means $f(n) = \Omega(g(n))$.
15. O and Ω — $\Theta(g(n))$ means both $O(g(n))$ and $\Omega(g(n))$.